

Grover's Search Algorithm

Amplify the answer you want. Build the oracle and the diffuser, then watch amplitude drain from every wrong state in $\sqrt{N}$ iterations.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

$N = 2^n$ items are indexed by n qubits, w is the marked item, $|s\rangle = H^{\otimes n}|0\rangle^{\otimes n}$ is the uniform superposition, and M is the number of marked items when there is more than one. Bit strings are written $q_{n-1} \cdots q_0$.

LESSON 7.1

Unstructured search

KEY IDEA

With no structure to exploit, a classical search must check items one by one. Grover's algorithm finds the marked item with about $\frac{\pi}{4}\sqrt{N}$ queries, and no quantum algorithm can do better than order $\sqrt{N}$.

$$f : \{0, \dots, N-1\} \rightarrow \{0,1\}, \quad f(x) = 1 \Leftrightarrow x = w \quad (7.1.1)$$

the search problem, as an oracle

"Unstructured" means the only way to learn about an item is to ask the oracle about it. No sorting, no hashing, no gradient. Classically, each query rules out one item.

$$\mathbb{E}[Q_{\text{classical}}] = \frac{N+1}{2}, \quad Q_{\text{Grover}} \approx \frac{\pi}{4}\sqrt{N} \quad (7.1.2)$$

expected queries: opening boxes in turn, against Grover

ITEMS N	CLASSICAL, ON AVERAGE	CLASSICAL, WORST CASE	GROVER ITERATIONS
4	2.5	4	1
1024	512.5	1024	25
10^6	$\approx 5 \times 10^5$	10^6	785
10^{12}	$\approx 5 \times 10^{11}$	10^{12}	785 398

The speed-up is quadratic, and it is optimal. Any quantum algorithm that finds w with high probability must query the oracle order $\sqrt{N}$ times. This lower bound (Bennett, Bernstein, Brassard and Vazirani) is why search is not expected to become exponentially faster.

$$Q_{\text{quantum}} = \Omega(\sqrt{N}) \quad (7.1.3)$$

the BBBV lower bound: Grover is optimal

DEFINITION · THE ORACLE IS THE BOTTLENECK

Counting queries treats the oracle as free. In practice it is a reversible circuit that evaluates f , and its cost multiplies every iteration.

WORKED EXAMPLE · FOUR BOXES

Classically, open boxes in a random order: the prize is in box 1, 2, 3 or 4 of that order with probability $\frac{1}{4}$ each, so the expected number opened is $\frac{1}{4}(1 + 2 + 3 + 4) = 2.5$.

Grover's single iteration on two qubits finds it with probability 1, as the walkthrough on the algorithms page shows.

LESSON 7.2

Marking the answer

KEY IDEA

The oracle flips the sign of the marked item's amplitude and leaves everything else alone. The probabilities do not change, so marking alone reveals nothing.

$$O_w|x\rangle = (-1)^{f(x)}|x\rangle = \begin{cases} -|x\rangle & x = w \\ |x\rangle & x \neq w \end{cases} \quad (7.2.1)$$

the phase oracle

$$O_w = I - 2|w\rangle\langle w| \quad (7.2.2)$$

the same oracle, as an operator: a reflection that reverses $|w\rangle$

Start in the uniform superposition, where every amplitude is $1/\sqrt{N}$. After the oracle one amplitude is $-1/\sqrt{N}$, and every probability is still $1/N$.

$$O_w|s\rangle = \frac{1}{\sqrt{N}} \sum_{x \neq w} |x\rangle - \frac{1}{\sqrt{N}} |w\rangle \quad (7.2.3)$$

after marking: a sign, not a probability

A phase oracle can be built from a bit oracle with an ancilla in $|-\rangle$ (lesson 6.3), or directly from a multi-controlled Z . $C^{n-1}Z$ flips the sign of $|1 \cdots 1\rangle$ only. To mark some other string, wrap it in X gates on the qubits where w has a 0.

$$O_w = X^{\bar{w}} (C^{n-1}Z) X^{\bar{w}}, \quad X^{\bar{w}} = \bigotimes_{k: w_k = 0} X_k \quad (7.2.4)$$

marking any string with one multi-controlled Z

$$CZ = (I \otimes H) \text{CNOT} (I \otimes H) \quad (7.2.5)$$

for two qubits: a controlled- Z from a CNOT between Hadamards

MARKED $w = q_1q_0$

ORACLE ON TWO QUBITS

11	CZ
10	X on q_0 , CZ, X on q_0
01	X on q_1 , CZ, X on q_1
00	X on both, CZ, X on both

WORKED EXAMPLE · TWO QUBITS, $w = 11$

$|s\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. CZ gives $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$.

Each outcome still has probability $\frac{1}{4}$: measuring now would be a blind guess. The walkthrough draws exactly this, one bar flipping below the axis.

LESSON 7.3

Amplitude amplification

KEY IDEA

Everything happens in a plane spanned by $|w\rangle$ and the uniform superposition of the other items. The oracle and the diffuser are two reflections in that plane, and two reflections make a rotation towards $|w\rangle$.

$$|s\rangle = \sin \theta |w\rangle + \cos \theta |s'\rangle, \quad \sin \theta = \frac{1}{\sqrt{N}} \quad (7.3.1)$$

the start, in the plane of $|w\rangle$ and $|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$

The starting state lies at a small angle θ above $|s'\rangle$. The oracle reverses the $|w\rangle$ component, which reflects the state about $|s'\rangle$. The diffuser reflects about $|s\rangle$ itself.

$$O_w = I - 2|w\rangle\langle w|, \quad D = 2|s\rangle\langle s| - I \quad (7.3.2)$$

the two reflections

Composing two reflections whose mirrors meet at angle θ rotates by 2θ . Each Grover iteration therefore moves the state 2θ closer to $|w\rangle$, and after k iterations it has turned through $2k\theta$ from its start.

$$G = D O_w, \quad G^k |s\rangle = \sin((2k+1)\theta) |w\rangle + \cos((2k+1)\theta) |s'\rangle \quad (7.3.3)$$

k Grover iterations

$$P_k = \sin^2((2k+1)\theta) \quad (7.3.4)$$

the probability of measuring w after k iterations

STATE	ANGLE ABOVE $ s'\rangle$	AMPLITUDE ON $ w\rangle$
$ s\rangle$	θ	$\sin \theta$
$O_w s\rangle$	$-\theta$	$-\sin \theta$
$D O_w s\rangle = G s\rangle$	3θ	$\sin 3\theta$
$G^k s\rangle$	$(2k+1)\theta$	$\sin((2k+1)\theta)$

The same analysis works whenever a procedure succeeds with probability $p = \sin^2 \theta$. That generalisation is *amplitude amplification*, the subject of lesson 7.7.

WORKED EXAMPLE · THE ANGLE FOR FOUR ITEMS

$N = 4$: $\sin \theta = \frac{1}{2}$, so $\theta = 30^\circ$. The oracle takes the state to -30° , and the diffuser reflects it about the 30° line to 90° : exactly $|w\rangle$. One iteration, certainty.

LESSON 7.4

The diffuser

KEY IDEA

The diffuser replaces each amplitude a by $2\bar{a} - a$, a reflection about the mean. Amplitudes near the mean barely move, while one far below it is thrown far above.

$$D = 2|s\rangle\langle s| - I = H^{\otimes n} (2|0\rangle\langle 0| - I) H^{\otimes n} \quad (7.4.1)$$

the diffuser, and why it is built from Hadamards

$$D \sum_x a_x |x\rangle = \sum_x (2\bar{a} - a_x) |x\rangle, \quad \bar{a} = \frac{1}{N} \sum_x a_x \quad (7.4.2)$$

inversion about the mean

The second form follows in one line: $2|s\rangle\langle s|\psi\rangle$ equals $2 \cdot \frac{1}{\sqrt{N}} \sum_x |x\rangle \cdot \frac{1}{\sqrt{N}} \sum_y a_y = 2\bar{a} \sum_x |x\rangle$, and subtracting $|\psi\rangle$ leaves $2\bar{a} - a_x$ on each basis state.

In a circuit, the reflection about $|0 \cdots 0\rangle$ is built from X gates and a multi-controlled Z . Built that way it comes out with an overall minus sign, which is a global phase and harmless.

$$X^{\otimes n} (C^{n-1} Z) X^{\otimes n} = I - 2|0 \cdots 0\rangle\langle 0 \cdots 0| = -(2|0 \cdots 0\rangle\langle 0 \cdots 0| - I) \quad (7.4.3)$$

the circuit implements $-D$

THE TWO-QUBIT DIFFUSER

q0: —H—X—●—X—H—
q1: —H—X—Z—X—H—

ITEM	AFTER THE ORACLE	MEAN $\bar{a}$	$2\bar{a} - a$
$ 00\rangle$	$+\frac{1}{2}$	$\frac{1}{4}$	0
$ 01\rangle$	$+\frac{1}{2}$	$\frac{1}{4}$	0
$ 10\rangle$	$+\frac{1}{2}$	$\frac{1}{4}$	0
$ 11\rangle$ (marked)	$-\frac{1}{2}$	$\frac{1}{4}$	+1

WORKED EXAMPLE · WHY THE MARKED ONE GROWS

For large N the mean is almost $1/\sqrt{N}$. An unmarked amplitude $a = 1/\sqrt{N}$ becomes $2\bar{a} - a \approx 1/\sqrt{N}$, almost unchanged.

The marked amplitude $-1/\sqrt{N}$ becomes about $3/\sqrt{N}$, three times its original size, while each unmarked one shrinks only slightly to keep the total probability at one.

LESSON 7.5

One iteration

KEY IDEA

One Grover iteration is the oracle followed by the diffuser. It costs one query and rotates the state by $2\theta \approx 2/\sqrt{N}$ towards the answer.

$$G = D O_w$$

(7.5.1)

one Grover iteration: mark, then reflect about the mean

ONE ITERATION ON TWO QUBITS, $N = 11$: THE CZ ORACLE, THEN THE DIFFUSER

q0: —H—●—H—X—●—X—H—M
q1: —H—Z—H—X—Z—X—H—M

For large N , $\theta \approx 1/\sqrt{N}$, so each iteration adds about $2/\sqrt{N}$ to the angle. The marked amplitude, $\sin((2k+1)\theta)$, grows almost linearly at first.

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta \approx \frac{3}{\sqrt{N}}$$

(7.5.2)

the marked amplitude after one iteration: roughly tripled

$$\theta = \arcsin \frac{1}{\sqrt{N}} \approx \frac{1}{\sqrt{N}} \quad (N \gg 1)$$

(7.5.3)

the rotation per half-step

N	θ	P BEFORE	P AFTER ONE ITERATION
4	30°	0.25	1.000
16	14.48°	0.0625	0.473
64	7.18°	0.0156	0.135
1024	1.79°	0.00098	0.0088

The cost of an iteration is one oracle call plus the diffuser, whose multi-controlled Z needs $O(n)$ elementary gates with an ancilla. Total gates therefore grow as $O(\sqrt{N} \cdot n)$ plus the oracle's own cost.

$$\text{gates} = O\left(\sqrt{N} (n + c_{\text{oracle}})\right)$$

(7.5.4)

the full circuit's size

WORKED EXAMPLE · FOUR ITEMS, EXACTLY

$N = 4$, $\theta = \frac{\pi}{6}$. After one iteration the angle is $3\theta = \frac{\pi}{2}$, so $P_1 = \sin^2 \frac{\pi}{2} = 1$.

Amplitude by amplitude, the table of lesson 7.4 shows the same thing: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}) \mapsto (0, 0, 0, 1)$. Two qubits and one query find the item with certainty.

LESSON 7.6

How many iterations

KEY IDEA

Stop when the state is closest to $|w\rangle$, after about $\frac{\pi}{4}\sqrt{N}$ iterations. The state keeps rotating, so running longer moves it past the answer and the success probability falls again.

$$k^* = \left\lfloor \frac{\pi}{4\theta} \right\rfloor \approx \frac{\pi}{4}\sqrt{N} \quad (7.6.1)$$

the optimal number of iterations

$$P_{k^*} \geq 1 - \frac{1}{N} \quad (7.6.2)$$

the success probability at the optimum

The optimum comes from asking for $(2k+1)\theta \approx \frac{\pi}{2}$. Since k must be a whole number, the final angle misses $\frac{\pi}{2}$ by at most θ , which costs at most $\sin^2 \theta = 1/N$ in probability.

N	θ	k^*	P_{k^*}
4	30.00°	1	1.0000
16	14.48°	3	0.9613
64	7.18°	6	0.9966
1024	1.79°	25	0.9995

Overshooting is not harmless. The probability oscillates with period about $\pi/(2\theta)$ iterations, so doubling the optimal count brings it back near zero.

$$N = 4, k = 2: \quad P_2 = \sin^2 \frac{5\pi}{6} = \frac{1}{4} \quad (7.6.3)$$

one iteration too many, and four items are back to a guess

With M marked items the angle is larger and fewer iterations are needed. When M is unknown, *quantum counting* estimates it first, or a randomised schedule of iteration counts finds a marked item in expected $O(\sqrt{N/M})$ queries anyway.

$$\sin \theta = \sqrt{\frac{M}{N}}, \quad k^* \approx \frac{\pi}{4} \sqrt{\frac{N}{M}} \quad (7.6.4)$$

M marked items

WORKED EXAMPLE · A MILLION ITEMS

$N = 2^{20} = 1\,048\,576$; $\theta = \arcsin(1/1024) \approx 0.05596^\circ$, so $k^* = \lfloor \pi/(4\theta) \rfloor = 804$ iterations, and $P_{804} > 0.9999$.

A classical search would open about half a million boxes on average.

LESSON 7.7

What a quadratic speed-up buys

KEY IDEA

Grover turns N into $\sqrt{N}$. That halves the exponent of brute-force search, which is significant, but it is countered simply by doubling key lengths, unlike Shor's exponential speed-up.

brute-force search over k -bit keys: $2^k \rightarrow \approx 2^{k/2}$ oracle calls

(7.7.1)

Grover against symmetric keys

TARGET	CLASSICAL	WITH GROVER	REMEDY
AES-128 key	2^{128}	$\approx 2^{64}$	use AES-256
AES-256 key	2^{256}	$\approx 2^{128}$	already sufficient
n -bit hash preimage	2^n	$\approx 2^{n/2}$	longer outputs

The practical gain is smaller than the exponent suggests. Every iteration is a full reversible evaluation of the cipher, run under error correction, and the iterations must happen in sequence. Splitting the search across p machines helps only by $\sqrt{p}$.

$$p \text{ machines, each searching } N/p \text{ items: } \approx \frac{\pi}{4} \sqrt{\frac{N}{p}} \text{ iterations each}$$

(7.7.2)

parallel Grover: total work grows as $\sqrt{pN}$

The idea is more general than search. Any procedure that succeeds with probability p , without telling you whether it did, would classically be repeated about $1/p$ times. *Amplitude amplification* reaches constant success in about $1/\sqrt{p}$ repetitions, the same quadratic saving.

repetitions: $O\left(\frac{1}{p}\right) \rightarrow O\left(\frac{1}{\sqrt{p}}\right)$

(7.7.3)

amplitude amplification

SPEED-UP	EXAMPLE	EFFECT ON CRYPTOGRAPHY
quadratic	Grover, amplitude amplification	double symmetric key sizes
exponential	Shor (module 8)	RSA and elliptic curves broken outright

WORKED EXAMPLE · HOW LONG 2^{64} TAKES

Suppose, optimistically, a fault-tolerant machine completes one AES Grover iteration per microsecond. $2^{64} \approx 1.8 \times 10^{19}$ iterations take 1.8×10^{13} seconds, about 580 000 years, in sequence. Classical brute force of AES-128 is hopeless too, but that figure shows why doubling the key length is the accepted answer.

MODULE 7 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

the search problem, as an oracle	$f : \{0, \dots, N-1\} \rightarrow \{0,1\}, \quad f(x) = 1 \Leftrightarrow x = w$	(7.1.1)
expected queries: opening boxes in turn, against Grover	$\mathbb{E}[Q_{\text{classical}}] = \frac{N+1}{2}, \quad Q_{\text{Grover}} \approx \frac{\pi}{4}\sqrt{N}$	(7.1.2)
the BBBV lower bound: Grover is optimal	$Q_{\text{quantum}} = \Omega(\sqrt{N})$	(7.1.3)
the phase oracle	$O_w x\rangle = (-1)^{f(x)} x\rangle = \begin{cases} - x\rangle & x = w \\ x\rangle & x \neq w \end{cases}$	(7.2.1)
the same oracle, as an operator: a reflection that reverses $ w\rangle$	$O_w = I - 2 w\rangle\langle w $	(7.2.2)
marking any string with one multi-controlled Z	$O_w = X^{\bar{w}} (C^{n-1}Z) X^{\bar{w}}, \quad X^{\bar{w}} = \bigotimes_{k: w_k=0} X_k$	(7.2.4)
the start, in the plane of $ w\rangle$ and $ s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} x\rangle$	$ s\rangle = \sin \theta w\rangle + \cos \theta s'\rangle, \quad \sin \theta = \frac{1}{\sqrt{N}}$	(7.3.1)
the two reflections	$O_w = I - 2 w\rangle\langle w , \quad D = 2 s\rangle\langle s - I$	(7.3.2)
k Grover iterations	$G = D O_w, \quad G^k s\rangle = \sin((2k+1)\theta) w\rangle + \cos((2k+1)\theta) s'\rangle$	(7.3.3)
the probability of measuring w after k iterations	$P_k = \sin^2((2k+1)\theta)$	(7.3.4)
the diffuser, and why it is built from Hadamards	$D = 2 s\rangle\langle s - I = H^{\otimes n} (2 0\rangle\langle 0 - I) H^{\otimes n}$	(7.4.1)
inversion about the mean	$D \sum_x a_x x\rangle = \sum_x (2\bar{a} - a_x) x\rangle, \quad \bar{a} = \frac{1}{N} \sum_x a_x$	(7.4.2)
one Grover iteration: mark, then reflect about the mean	$G = D O_w$	(7.5.1)
the marked amplitude after one iteration: roughly tripled	$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta \approx \frac{3}{\sqrt{N}}$	(7.5.2)
the optimal number of iterations	$k^* = \left\lceil \frac{\pi}{4\theta} \right\rceil \approx \frac{\pi}{4}\sqrt{N}$	(7.6.1)
the success probability at the optimum	$P_{k^*} \geq 1 - \frac{1}{N}$	(7.6.2)
M marked items	$\sin \theta = \sqrt{\frac{M}{N}}, \quad k^* \approx \frac{\pi}{4} \sqrt{\frac{N}{M}}$	(7.6.4)
Grover against symmetric keys	brute-force search over k -bit keys: $2^k \rightarrow \approx 2^{k/2}$ oracle calls	(7.7.1)
parallel Grover: total work grows as $\sqrt{pN}$	p machines, each searching N/p items: $\approx \frac{\pi}{4} \sqrt{\frac{N}{p}}$ iterations each	(7.7.2)

amplitude amplification

$$\text{repetitions: } o\left(\frac{1}{p}\right) \rightarrow o\left(\frac{1}{\sqrt{p}}\right)$$

(7.7.3)